

Design Science Research Evaluation

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Abstract. The consensus view is that the rigorous evaluation of design science (DS) artifacts is essential. There are many types of DS artifacts and many forms of evaluation; what is missing is guidance for how to perform the evaluation, more specifically, what evaluation methods to use with specific DS research outputs. Here we find and review 148 DS research articles published in a selected set of information systems (IS), computer science (CS) and engineering journals. We analyze the articles to develop taxonomies of DS artifact types and artifact evaluation methods; we apply these taxonomies to determine which evaluation methods are associated in the literature with particular artifacts. We show that there are several popular “artifact - evaluation method” combinations in the literature. The results inform DS researchers of usual and customary combinations of research artifacts and evaluation methods, potentially providing them with rationale and justification for an evaluation method selection.

1 Introduction

The importance of evaluating DS research artifacts is well supported in the literature [1, 2]. These articles provide justification for the publication of DS research efforts and outcomes in quality information systems (IS) research outlets, provided that the artifacts are, in addition to other attributes, rigorously evaluated. That such evaluation is an essential component of a DS research contribution is emphasized as “crucial” in Hevner et al. [2] and in Peffers et al. [1], as well as in discussions about how to conduct DS research, e.g., [3-5]. Other researchers have also emphasized the importance of evaluation, e.g., [6-11].

Notwithstanding the consensus that evaluation is essential, there is little guidance about what is desirable, acceptable or customary in evaluation. Artifacts should be evaluated with criteria based on the requirements of the context in which the artifact is implemented, according to Hevner et al. [2]; for example, in terms such as “functionality, completeness, consistency, accuracy, performance, reliability, usability, fit with the organization and other relevant quality attributes.” Evaluation methods might include case studies, field studies, static analysis, architectural analysis, optimization, dynamic analysis, controlled experiments, simulation, functional testing, structural

testing, informed argument, or scenarios [2, p. 86, Table 2.]. Peffers et al. [1] advocated “observ[ing] how well the artifact supports a solution to the problem.” Case studies reported in Peffers et al [1] were evaluated using field studies, observation of reuse rates, performance testing, and client feedback about usefulness.

How are DS research artifacts evaluated? There is considerable variety in the artifacts that result from DS research efforts. Are there, consequently, characteristics of these artifacts that tend to lead to certain kinds of evaluation? These are important questions because the evaluation and presentation of DS research outcomes is essential to the efficacy of DSR research, as well as to the professional success of researchers. DS research papers are unlikely to be published in influential outlets, unless authors can make persuasive arguments that artifacts were appropriately evaluated.

In this paper, we locate and examine DS research papers from IS and from engineering research journals. We classify the papers to create taxonomies of the DS research artifact types and of the evaluation method types in use. Using these classifications, we observe associations between types of artifacts and the methods used to evaluate them, showing which evaluation methods have been used to evaluate which types of artifacts. We then develop case studies of exemplary studies to show how the artifacts have been evaluated.

2 Data

The data consists of published DS research papers in major IS and engineering journals. We selected ten well-regarded journals that we judged were likely to publish some design science articles (Table 1). Since DS researchers in the IS discipline have traditionally published at least some of their research in engineering journals, we included those engineering journals that are most important to the IS discipline as well. Five selected journals are in the core MIS discipline and five are in computer science and engineering. All of them are listed in prominent positions on the AIS Journal ranking lists [12].

From each of these journals we identified up to 25 articles, searching up to five volumes and stopping when we had identified 25 articles or completed five volumes, whichever occurred first. One researcher included all articles that he classified as design science into the candidate set of articles, taking an inclusive approach; including an article when in doubt, rather than omitting one. This resulted in a total of 159 articles.

Two additional researchers independently audited each candidate article to determine whether it was correctly classified as DS research. Each researcher identified articles for which the classification as design science was not clear. The two auditing researchers discussed the doubtfully classified articles and made a decision about whether to include the article by consensus. This process resulted in the removal of eleven articles from the study because the auditors agreed that they were not DS research. The auditing step ensured that all papers entered into the study were actually DS research papers. This resulted in a sample of 148 articles. As we expected more DS articles appeared in computer science and engineering journals than in IS journals,

as DS is a more prevailing research approach in those disciplines. The final total included 117 articles in computer science and engineering journals and 31 in IS journals.

Table 1. Information systems and engineering journals included in search

Journal Name	Acronym
ACM Transactions on Database Systems	ACM TDB
European Journal of Information Systems	EJIS
IEEE Transactions on Engineering Management	IEEE TEM
IEEE Transactions on Knowledge and Data Engineering	IEEE TKDE
IEEE Transactions on Software Engineering	IEEE TSE
IEEE Transactions on Systems, Man & Cybernetic	IEEE TSMC
Information and Management	I&M
Information Systems Research	ISR
Journal of Management Information Systems	JMIS
MIS Quarterly	MISQ

3 Analysis

We classified the papers by artifact type and evaluation method. The two researchers independently evaluated the content of each paper for these two characteristics and established item categories in the two dimensions, in order to develop sets of artifact types and evaluation methods. We used an open coding approach; two researchers independently categorized the papers, defining and naming their own categories without an established list. After the independent assessments, the two researchers compared their results; 91 percent of the artifact classifications and 81 percent of the method classifications were consistent between the two researchers. Different category names with the same meaning were consolidated into one category name without counting that as a difference. To resolve the remaining classification differences, the two researchers reread the respective articles and obtained consensus in a discussion in which they explained their reasoning to each other. Differences were resolved in an iteration of this process and the codes were consolidated into six artifact types and eight evaluation method types. At the same time they established the artifact and evaluation method labels and definitions, checking paper content to ensure that the resulting categorizations were consistent and that the definitions were consistently exclusive.

The artifact classification includes conceptual artifacts like constructs, models, and frameworks, as well as methods, which are conceptual actionable instructions. Formal

logical instructions are classified as algorithms and actual hardware or software implementations are classified as instantiations (Table 2).

Table 2. Artifact Types

Algorithm	An approach, method, or process described largely by a set of formal logical instructions.
Construct	Concept, assertion, or syntax that has been constructed from a set of statements, assertions, or other concepts.
Framework	Meta-model
Instantiation	The structure and organization of a system’s hardware or system software or part thereof.
Method	Actionable instructions that are conceptual (not algorithmic)
Model	Simplified representation of reality documented using a formal notation or language.

The evaluation method classification includes technical experiments as a means to evaluate the performance of an artifact without the involvement of research subjects, and subject-based experiments that use research subjects to assess the validity of the assertions that motivated the development of the artifact are true. The classification further differentiates between prototypes, illustrative scenarios, case studies, and action research. Prototypes are the implementation of artifacts to demonstrate their utility; illustrative scenarios apply the artifact in a synthetic or real world situation to demonstrate its utility; and case studies implement the artifact in a real-world situation to evaluate not only its utility, but also its effect on its environment; action research also implements the artifact in a real-world situation to evaluate its effect on the environment, but does that in the context of a research intervention. Logical arguments and expert evaluations are also part of the evaluation method classifications (Table 3).

3.1 Clusters of artifacts and evaluation methods

Table 4 shows the results of our analysis, in terms of the count of the number of papers at the intersection of each artifact and evaluation method. It shows that, in this sample of papers, algorithms, methods, and models are the most popular artifact types across all journals; however, the other artifact types were also well represented. Technical experiments and illustrative scenarios are the most commonly used evaluation methods.

To differentiate between IS vs. CS and engineering DS research, we split the results from Table 4 into Table 5 for IS journals and Table 6 for CS and engineering

journals. Since the majority of DS research papers in our sample appeared in the CS and engineering journals, the results for those journals fairly closely mirrored those for the whole sample. The results for the IS journals reflected the much smaller representation of DS research in those journals. Nonetheless, despite the small numbers, discernible patterns of association appeared. In particular, case studies and illustrative scenarios figured more predominantly in the counts of evaluation methods. The choices of artifact type appeared to be much more evenly distributed in the IS journals than they were in the CS and engineering journals.

Table 3. Evaluation Method Types

Logical Argument	An argument with face validity.
Expert Evaluation	Assessment of an artifact by one or more experts (e.g., Delphi study).
Technical Experiment	A performance evaluation of an algorithm implementation using real-world data, synthetic data, or no data, designed to evaluate the technical performance, rather than its performance in relation to the real world.
Subject-based Experiment	A test involving subjects to evaluate whether an assertion is true.
Action Research	Use of an artifact in a real-world situation as part of a research intervention, evaluating its effect on the real-world situation.
Prototype	Implementation of an artifact aimed at demonstrating the utility or suitability of the artifact.
Case Study	Application of an artifact to a real-world situation, evaluating its effect on the real-world situation.
Illustrative Scenario	Application of an artifact to a synthetic or real-world situation aimed at illustrating suitability or utility of the artifact.

As one might expect, the choice of evaluation methods is driven by choice of artifacts; particular artifacts lend themselves to evaluation with particular methods. To analyze these clusters, we created groups of articles employing unique artifact type / evaluation method combinations. Tables 4 through 6 show the number of articles that belong to each artifact/evaluation method pair. The tables demonstrate that certain artifact / evaluation method combinations occur far more frequently than others. For example, in IS journals, methods are most frequently evaluated using case studies, while in computer science and engineering journals, methods are most commonly evaluated using technical experiments. In both disciplines algorithms are most frequently evaluated using technical experiments.

Table 4. Distribution of Evaluation Methods by Artifact Type (all journals)

	Logical Argument	Expert Evaluation	Technical Experiment	Subject-Based Experiment	Prototype	Action Research	Case Study	Illustrative Scenario	none	Total
Algorithm	1		60	1				3		65
Construct	3		3	2	2			2		12
Framework	1	1			1		1	4	1	9
Instantiation			5	1	1			1		8
Method	2		14	4			7	6		33
Model	3		10		2	2		4		21
Total	10	1	92	8	6	2	8	20	1	

Table 5. Distribution of Evaluation Methods by Artifact Type (IS journals)
[EJIS, I&M, ISR, JMIS, MISQ]

	Logical Argument	Expert Evaluation	Technical Experiment	Subject-Based Experiment	Prototype	Action Research	Case Study	Illustrative Scenario	none	Total
Algorithm			4							4
Construct	1				1					2
Framework	1	1					1	1	1	5
Instantiation			3					1		4
Method	1		2	2			6	1		12
Model			1		1	1		1		4
Total	3	1	10	2	2	1	7	4	1	

Table 6. Distribution of Evaluation Methods by Artifact Type (CS/Eng journals) [ACM TDB, IEEE TEM, IEEE TKDE, IEEE TSWE, IEEE TSMC]

	Logical Argument	Expert Evaluation	Technical Experiment	Subject-Based Experiment	Prototype	Action Research	Case Study	Illustrative Scenario	none	Total
Algorithm	1		56	1				3		61
Construct	2		3	2	1			2		10
Framework					1			3		4
Instantiation			2	1	1					4
Method	1		12	2			1	5		21
Model	3		9		1	1		3		17
Total	7	0	80	6	4	1	1	16	0	

3.2 Exemplar Papers

For each research artifact / evaluation method cluster, we identified an exemplar paper for further analysis. The objective in this selection was to obtain a typical representative for each included specific artifact / evaluation method pair. This is consistent with a theoretical sampling approach in qualitative research, treating the papers as cases to be entered into further analysis. This final analysis step examines the reasoning provided in the exemplar papers for the choice of evaluation method.

4 Two Exemplar Studies

Here we examine the reasoning provided in the exemplar papers for the choice of evaluation method. We believe that artifact type, context, and data availability are issues that contribute to the selection of an evaluation method, but we expect that a more refined set of criteria will emerge from the analysis. Because space in this publication is limited, we present here just two of the selected exemplar papers.

4.1 Exemplar Paper 1: Instantiation evaluated by prototype

Chen et al (2003) designed a flexible database system (FDS) to provide superior query performance in production systems, e.g., that of an airline or large retailer, where

the database is very large and the number of read queries is much larger (perhaps 1000 times larger) than that of write queries and where read queries are very diverse, often requiring joins among diverse tables.

To achieve improved performance the FDS employs a number of read-only, de-normalized database systems that are each intended to field queries of different characteristics. The FDS consists of a query analyzer, DBS_0 (a normalized database system), $DBS_1 - DBS_n$ (de-normalized read-only systems), a scheduler, and a performance monitor.

Designed artifact: system instantiation.

We classified the designed artifact as an instantiation because it represents in idea or concept that is or is intendedly expressed by an example system, rather than, for example, in a modeling language. The query analyzer receives a queries from system users and assigns each to one of the DBS, such that if the query is an update it is assigned to DBS_0 or if the query is a read-only it is assigned to one of $DBS_1 - DBS_n$, depending on which DBS is predicted to produce the best performance for the query.

The design of the FDS follows the steps:

1. Design of the update data structure, DBS_0 , using a conventional database design approach.
2. Design of the read-only candidate structures, scaling down from an exhaustive set of de-normalized designs through use of expert database knowledge to a feasible set.
3. Build a knowledge base of queries to determine which DBS provide the fastest time for queries with given characteristics.
4. Decide which read-only DBS to keep active.
5. Assign queries to the most appropriate structure.
6. Determine conditions to refresh the read-only DBS from DBS_0 .
7. Restructure the FDS when the distribution of query types warrant it.

Evaluation: prototype.

Chen et al [13] develop two prototype FDS, one of which is based on a very simple data structure and the other a more complex, production level system, with seven tables, each with 10 to 40 attributes, and from 625 to 6250 tuples, and they use them to demonstrate six of the seven steps above. Experiments using the two prototypes show an average of 40% less processing times can be achieved using the FDS, compared with a 3NF DBS. They estimate that this savings is easily sufficient to offset the cost of additional storage, refresh and restructuring required to implement and maintain the FDS.

The use of a prototype instantiation to demonstrate the efficacy of a design can provide strong evidence when used to show that a design works as intended, is useful for its intended purpose, or has the potential to achieve an expected performance level. The use of a specific instantiation suggests that the artifact should be evaluated on its directly observable performance, e.g., processing time.

4.2 Exemplar Paper 2: Method evaluated by case study.

Peppers, Gengler, and Tuunanen [14] designed a method intended to facilitate wide participation in data collection for IS planning activities, while maintaining a focus on ideas that are important for the firm. Wide spread participation should allow the firm to take advantage of knowledge about the value of potential systems that is widely dispersed among people in and around the firm, rather than only the knowledge that is held among a small circle of executives. The method needed, however, to overcome the limitation of traditional bottom-up planning processes, where the participants overwhelm managers with many self-serving ideas that are difficult to evaluate, but are mostly valueless.

The method is based on personal construct theory [15]. PCT holds that every observer has models for how the universe works, such that states of the universe have consequences and these consequences impact the values of the observer. PCT is extended in this research to a designed method for data collection, analysis, and ideation that can produce ideas for potentially valuable new systems for the organization.

Designed artifact: method.

The designed artifact, critical success chains (CSC), consists of a set of procedures that informs the beginning of the IS systems planning process, from initial data collection, to analysis, and to ideation, to produce a set of feasible ideas for new systems. We classify this as a method because the instructions are actionable and conceptual.

CSC proceeds in a series of four phased activities:

- pre-study (scope the study, determine sample participant, gather stimuli),
- discovery (data collection through intensive one-on-one interviews),
- analysis (aggregating and modeling the data),
- integration (transforming the modeled ideas into feasible system ideas)

Evaluation: case study.

Peppers, Gengler, and Tuunanen [14] demonstrated the CSC method in two case studies, an initial exploratory case study at Rutgers University and a more extensive and thorough study at Digia, a Helsinki software development firm. At Digia the scope of the study was ideas for ‘killer applications’ for mobile financial services. Data collection with a sample of 32 lead users and experts resulted in 147 chains, or arguments for specific system functionality. Analysis reduced this data to five graphical network models, each describing functionality, reasoning, and value, for a product bundle. An ideation workshop used these results to develop feasible ideas for three new products that the firm intended to explore further.

We classified this evaluation as a case study because the method was demonstrated in the context of use to affect real phenomena in the organization. The case study lends itself for use in evaluating the efficacy of a designed object that is intended to be used in a complex organizational setting where a simple experiment or other simple test could not be used to adequately show the efficacy or performance of the object. Case studies can provide much stronger evidence of efficacy or performance than

simple arguments or illustrative scenarios. They vary greatly in the strength of evidence of efficacy or performance, often depending on how far the case carries the results of use of the method towards an impact on the firm. In this case the firm intended to explore the ideas further, however, the case study does not follow the story through to full execution, because, as is often the circumstance, full execution would occur well outside the desired time frame of the case story.

A potential weakness of case studies comes from the specificity of the context; the more idiosyncratic the case story, the less generalizable the evidence. While an illustrative scenario can be tailored to an ideal context, a case study, based as it must be on a real context, is likely to come with less than ideal observed facts.

5 Discussion

This research has led to the developed of a taxonomy of design science research artifacts and a taxonomy of artifact evaluation method types, based on use in publications in the IS and CS and engineering disciplines. Six artifact types and eight evaluation method types have emerged. The analysis of the publication count in each of these categories has shown that some artifact types and evaluation methods are more commonly used than others, that there are some differences between IS vs. CS and engineering publications with respect to artifact types and evaluation methods, and that there seem to be clusters of artifacts and evaluation among the papers.

Overall, technical experiments are dominant, representing a clear majority of the evaluations (92 of 148). Illustrative scenarios, with 20, follow this. Algorithms make up a near majority of the artifacts, followed by methods and models. When we look at the disciplines separately, the results do not differ much, except that for the papers published in IS journals there is a cluster of methods evaluated by case studies that represent a significant part of the instances.

An analysis of the artifact / evaluation method clusters was less conclusive: clear clusters are missing in the overall data set, except for the “algorithm - technical experiment” combination. The same holds for the IS article sample, however, to little surprise it was confirmed that IS researchers are more keen to use qualitative evaluation methods such as case studies and action research. Similarly CS and engineering researchers use often more technical evaluation methods for their research. Therefore, we can see more differences between the research disciplines. Nevertheless, the analysis has supported the notion that the artifact, as well as the culture of the discipline drive evaluation choices.

While certain artifacts are almost exclusively evaluated using a specific evaluation method (for example, algorithms are almost exclusively evaluated using technical experiments), other artifacts lend themselves to a choice with regards to the evaluation method selection. Methods are frequently evaluated using technical experiments, but also using case studies; CS and engineering outlets seem to favor the former, while IS outlets most frequently publish the latter. The analysis of the exemplar cases is shedding light on the motivation and the tradeoffs with regards to the evaluation method choice.

The Peffers et al. [14] evaluated a method using case studies; the discussion of this exemplar paper emphasized that the evaluation method chosen demonstrated the efficacy (performance) of the design artifact in an organizational context, however, the discussion also highlighted that a potential weakness of case study evaluation is their lack of generalizability. Technical experiments, which is the prevailing alternative evaluation method for methods, was evaluated in the context of another exemplar paper (not included in this version of the article for space considerations); this evaluation method would potentially address this weakness, as technical experiments can be designed for generalizability. However, this evaluation method demonstrates efficacy in a synthetic environment (its technical performance), and thus falls short of the case study in demonstrating performance in real world context. This discussion on the tradeoffs of evaluation choices for the method artifact is exemplary for several other artifacts that offer evaluation choices to the researcher. The extended version of the paper will include a larger set of exemplar papers that will help us to explore these choices in detail.

6 Conclusions, Limitations and Future Research

In this paper, we have reviewed a selection of journal articles, 148 in total, in order to assess what is the current practice of evaluating outcomes of DS research. More specifically, we were interested in understanding the connections between different artifacts and evaluation methods. The results of our study show some patterns of evaluation for DS artifacts and evaluation types in each of information systems and engineering. The patterns of evaluation potentially provide authors with examples and templates to support their own use of evaluation methods in future DS research.

As limitations to the study, we see that a more detailed analysis is needed. First of all, due space limitations we were not able to do detailed analysis of all combinations of artifacts and evaluation methods. In this paper, we do provide two exemplars that would represent the range of artifacts and evaluation. What remains to be done in this research is to examine the papers in this sample to understand the reasoning and tradeoffs behind the use of specific methods to evaluate specific artifacts. We plan to address this limitation in an extended version of this paper. A deep analysis of the reasoning and tradeoffs behind evaluation choices will, we hope, illuminate good practices. In addition, a more complete set of exemplar papers will help future DS researchers with a valuable set of examples to support good evaluation method choices and to help justify those choices.

It can be also argued that more comprehensive review of literature should be done in order to provide better overview of the matter in the literature. Although, we see that the current sample of articles is well justified as means for doing the analysis we do see that this argument can be made. For example, the bias in the current sample is towards more technical papers in software engineering. For this reason, the set of IS journals could be extended and journals that entertain publishing DS research outputs, such as Information & Management, Journal of Information Technology Theory and Application and Scandinavian Journal of Information Systems, could be included in

the review. There are also a number of DS research specific tracks in well respected IS conferences, which could be considered including the conference this paper has been written for. However, the current sample of articles has allowed us to develop the artifact and evaluation method taxonomies and helped us to identify the prevailing artifact / evaluation method combinations; this, in turn, has led to the identification of the exemplar articles that drove the analysis of the reasoning and tradeoffs of evaluation method choices. Thus, we consider that the sample is justified for the given purpose.

In the future, we should be able to develop heuristics for researchers to follow in order to choose the appropriate artifact / evaluation method combination for their research. Contingency theory based research in IS literature offers interesting examples of how to do this, see, e.g. [16]. However, we also recognize that the practitioner, in our case the researcher, should be involved in this research process as well. Therefore, this stream of research may call for more action research [see, e.g., 17], or perhaps action design style research [11].

7 References

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